T3 Field Sheet — Hull-Only Ballast Validation

Print this sheet and bring it to the test. Background, rationale, and pass conditions are in `tests.md`.

Pre-Requisites

- [ ] Sealed hull (both lids, syringe hole blocked — record method below)
- [ ] Kitchen scale accurate to 1 g
- [ ] Lead shot and a small weighing pot/cup
- [ ] Water container large enough to fully submerge the hull (250 mm × 110 mm)
- [ ] Something to dry the hull between steps
- [ ] Phone or audio recorder for observations *(or write in the Observations section)*

Syringe hole blocking method: ____________________________

Step 0 — Weigh the Empty Hull

Weigh the assembled, sealed, empty hull before adding any ballast.

Measurement	**Value**
Empty hull mass	_______ g
Predicted ballast (= 2376 – hull mass)	_______ g
Phase 1 start (= predicted – 500 g)	_______ g

Phase 1 — Coarse (250 g steps)

Start at the Phase 1 start value from Step 0. Add 250 g at each step. Stop when the hull sinks or is clearly at the transition.

Float state key: **F** = floats clearly / **N** = near waterline / **S** = sinks

Tilt key: **L** = level / **B** = bow up / **St** = stern up / **R** = rolled

Step	Total ballast (g)	Float (F/N/S)	Tilt (L/B/St/R)
1.1			
1.2			
1.3			
1.4			
1.5			
1.6			

Phase 1 bracket: floats at _______ g, sinks at _______ g

Phase 2 — Medium (50 g steps)

Start at the last clearly-floating mass from Phase 1. Add 50 g at a time.

Step	Total ballast (g)	Float (F/N/S)	Tilt (L/B/St/R)
2.1			
2.2			
2.3			
2.4			
2.5			
2.6			

Phase 2 bracket: floats at _______ g, sinks at _______ g

Phase 3 — Fine (10 g steps)

Start at the last clearly-floating mass from Phase 2. Add 10 g at a time.

Step	Total ballast (g)	Float (F/N/S)	Tilt (L/B/St/R)
3.1			
3.2			
3.3			
3.4			
3.5			
3.6			

Phase 3 bracket: floats at _______ g, sinks at _______ g

Phase 4 — Precision (1 g steps)

Start at the last clearly-floating mass from Phase 3. Add 1 g at a time.

Step	Total ballast (g)	Float (F/N/S)	Tilt (L/B/St/R)
4.1			
4.2			
4.3			
4.4			
4.5			
4.6			
4.7			
4.8			
4.9			
4.10			

Result

	Value
Last mass that clearly floats	_______ g
First mass that clearly sinks	_______ g
Empty hull mass (from Step 0)	_______ g
Implied displaced volume (= sink mass / 1.0)	_______ cm^3
Predicted displaced volume (from calc)	2376 cm^3
Difference	_______ cm^3

Observations

(Reference steps by number, e.g. 2.3. Leave blank if using audio recording.)

Spare Table A

Step	Total ballast (g)	Float (F/N/S)	Tilt (L/B/St/R)
A.1			
A.2			
A.3			
A.4			
A.5			
A.6			
A.7			
A.8			
A.9			
A.10			

Spare Table B

Step	Total ballast (g)	Float (F/N/S)	Tilt (L/B/St/R)
B.1			
B.2			
B.3			
B.4			
B.5			
B.6			
B.7			
B.8			
B.9			
B.10			